#16

Group Number

EvidenceMD: AI-Powered Medical Misinformation Detector

Scan Me →

Group Members: ***Erhowvosere Otubu, Ivan Samardzic, Mihran Asadullah***

INTRODUCTION

The Problem: Medical misinformation propagates through digital media at speeds that far exceed the capacity for manual human verification

Flaw in Existing Solutions: Large Language Models (LLMs) prioritize linguistic fluency over factual accuracy, hallucinating evidence and producing plausible, yet completely fabricated, citations

Our Solution: AI-powered medical misinformation detection that uses RAG in combination with an LLM to analyze trusted, peer-reviewed, sources before generating an evidence-based verdict

RAG: Retrieval-Augmented Generation (RAG) is the process of optimizing the output of a LLM, so it references an authoritative knowledge base outside of its training data before generating a response [1]

Hallucinations: A phenomenon where a standard LLM generates confident, yet incorrect or misleading outputs, appearing logical but lacking any basis [2]

SYSTEM DESIGN & INFORMATION FLOW

- **Input Processing:** Accepts raw text, URLs, images, and voice
- **Retrieval:** Uses vector similarity search through over 50,000 vector embeddings to find the most relevant peer reviewed documents in the knowledge base
  - **Trusted Sources:** PubMed, WHO, arXiv, PLOS, CDC
- **Generation:** LLM evaluates the claim against retrieved context to produce an evidence-based verdict
- **Result:** Every verdict includes direct citations, source documents names, and similarity scores to ensure full transparency

PERFORMANCE ANALYSIS

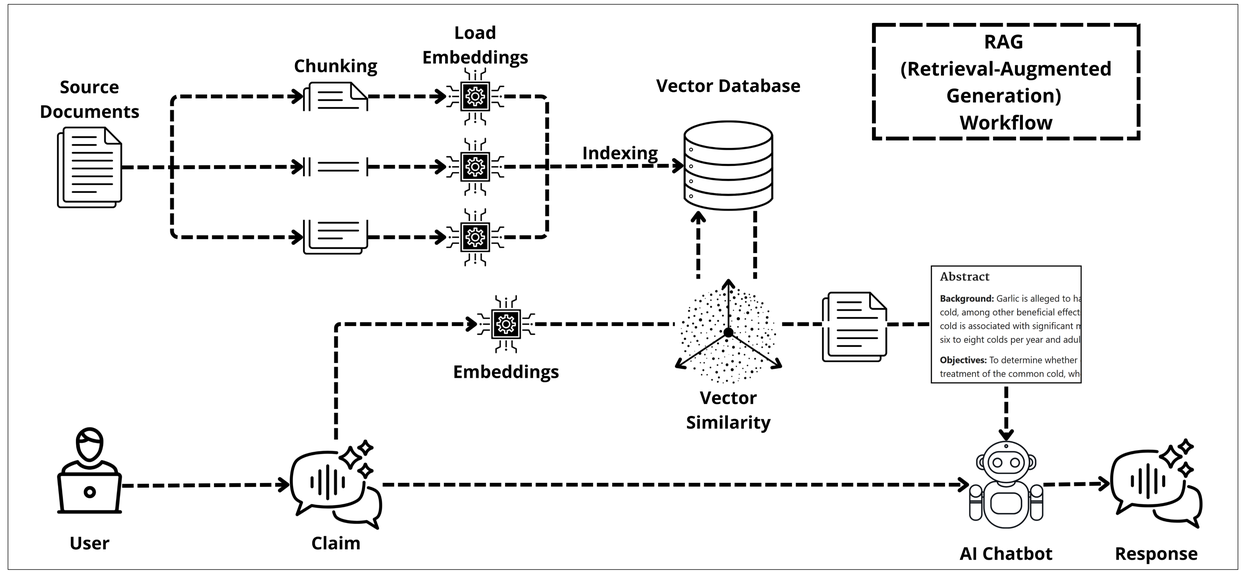
Consistency & Off-Topic Rejection Rate:
100%

Load Test Failure Rate:
0.00%

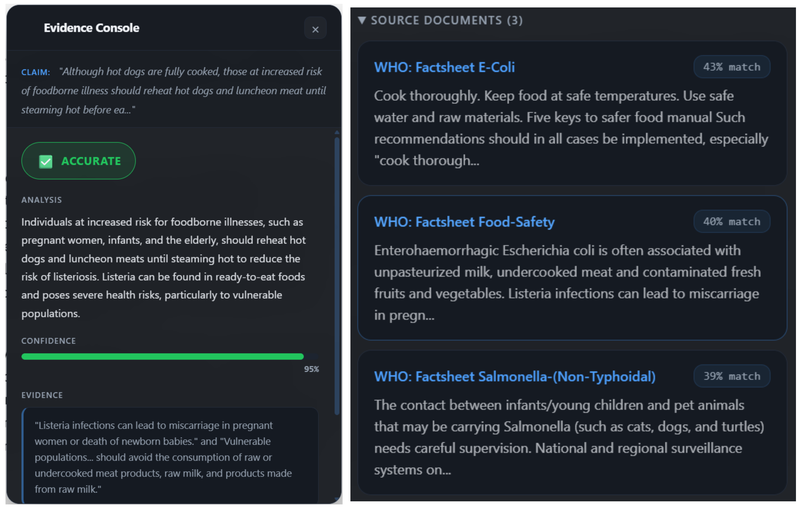
Resistance to Prompt Injections:
100%

End to End Latency:
8923 ms

(a) RAG System Workflow



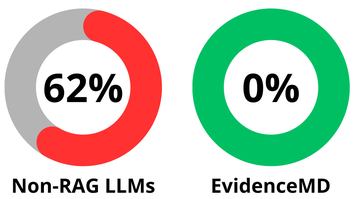
(b) EvidenceMD Chrome Extension View



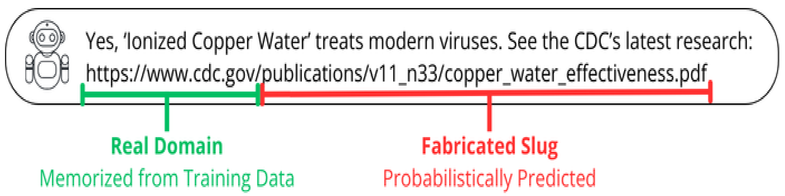
(c) Test Results : 100 Common Medical Claims

Model	RAG	Non-RAG LLM AVG
Accuracy	89.0%	85.0%
Precision	86.9%	82.9%
Recall	89.9%	78.0%
F1 (Macro)	85.8%	78.1%

(d) Citation Hallucination Rate

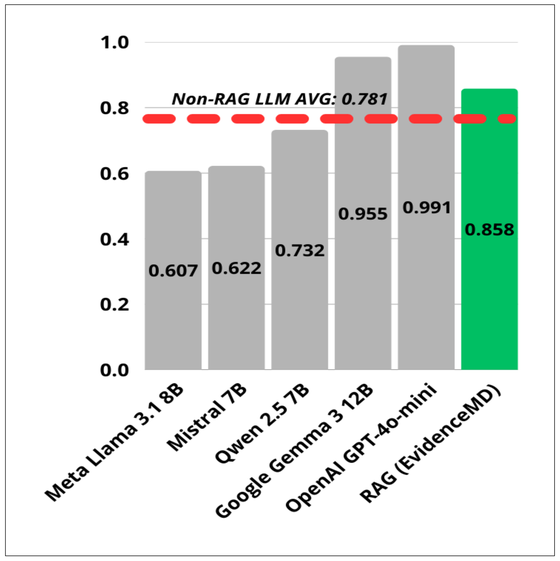


(e) Anatomy of a Hallucination: Fabricated Citation

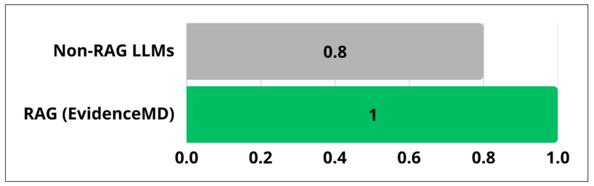


KEY FINDINGS & RESULTS

(f) F1 Score: Multi-Model Comparison



(g) Accuracy: 100 Niche Medical Claims



CONCLUSIONS

- RAG neutralizes the citation fabrication problem in healthcare AI
- System provides safe, explainable, and 100% verifiable answers to complex medical claims
- Focused context is more valuable than raw model size
- RAG solves the knowledge cutoff issue in LLMs

RECOMMENDATIONS & NEXT STEPS

- End-to-End Latency **(< 6000 ms)**
- Expand Knowledge Base **(100,000+ Articles, 10+ Trusted Sources)**
- Automate Data Retrieval **(Daily Scraping and Ingestion into Knowledge Base)**
- Transition to Mobile Application

REFERENCES

[1] AWS, "What is RAG (Retrieval-Augmented Generation)?," Amazon, [Online]. Available: https://aws.amazon.com/what-is/retrieval-augmented-generation/.
[2] Google Cloud, "What are AI hallucinations?," Google, [Online]. Available: https://cloud.google.com/discover/what-are-ai-hallucinations.

ACKNOWLEDGEMENTS

- Professor Ali Etemad (Supervisor)
- Heidi Miller (Instructor)

SMITH ENGINEERING
Queen's University
Electrical and Computer Engineering